51 Which of the following reactions will NOT give primary amine as the product?

(1) $CH_3\,CONH_2 \xrightarrow{Br_2\,/\,KOH}$ Product

(2) $CH_3CN \xrightarrow[(ii)\,H_3O\oplus]{(i)\,LiAlH_4}$ Product

(3) $CH_3NC \xrightarrow[(ii)\,H_3O\oplus]{(i)\,LiAlH_4}$ Product

(4) $CH_3CONH_2 \xrightarrow[(ii)\,H_3O\oplus]{(i)\,LiAlH_4}$ Product

52 Match **List - I** with **List - II** :

	List - I		**List - II**
A.	Coke	I.	Carbon atoms are sp^3 hybridised.
B.	Diamond	II.	Used as a dry lubricant
C.	Fullerene	III.	Used as a reducing agent
D.	Graphite	IV.	Cage like molecules

Choose the **correct** answer from the options given below :

(1) A-II, B-IV, C-I, D-III
(2) A-IV, B-I, C-II, D-III
(3) A-III, B-I, C-IV, D-II
(4) A-III, B-IV, C-I, D-II

53 Given below are two statements : one is labelled as **Assertion A** and the other is labelled as **Reason R** :

Assertion A : Metallic sodium dissolves in liquid ammonia giving a deep blue solution, which is paramagnetic.

Reasons R : The deep blue solution is due to the formation of amide.

In the light of the above statements, choose the **correct** answer from the options given below :

(1) Both **A** and **R** are true and **R** is the correct explanation of **A**.
(2) Both **A** and **R** are true but **R** is **NOT** the correct explanation of **A**.
(3) **A** is true but **R** is false.
(4) **A** is false but **R** is true.

54 In Lassaigne's extract of an organic compound, both nitrogen and sulphur are present, which gives blood red colour with Fe^{3+} due to the formation of -

(1) $Fe_4\left[Fe(CN)_6\right]_3 \cdot x\,H_2O$

(2) NaSCN

(3) $\left[Fe(CN)_5\,NOS\right]^{4-}$

(4) $\left[Fe(SCN)\right]^{2+}$

55 The conductivity of centimolar solution of KCl at 25°C is 0.0210 $ohm^{-1}\,cm^{-1}$ and the resistance of the cell containing the solution at 25°C is 60 ohm. The value of cell constant is -

(1) 1.34 cm^{-1} (2) 3.28 cm^{-1}
(3) 1.26 cm^{-1} (4) 3.34 cm^{-1}

56 Given below are two statements : one is labelled as **Assertion A** and the other is labelled as **Reason R** :

Assertion A : A reaction can have zero activation energy.

Reasons R : The minimum extra amount of energy absorbed by reactant molecules so that their energy becomes equal to threshold value, is called activation energy.

In the light of the above statements, choose the **correct** answer from the options given below :

(1) Both **A** and **R** are true and **R** is the correct explanation of **A**.
(2) Both **A** and **R** are true and **R** is **NOT** the correct explanation of **A**.
(3) **A** is true but **R** is false.
(4) **A** is false but **R** is true.

57 Which one is an example of heterogenous catalysis?

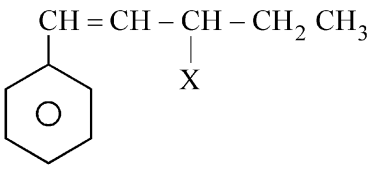
(1) Oxidation of sulphur dioxide into sulphur trioxide in the presence of oxides of nitrogen.

(2) Hydrolysis of sugar catalysed by H^+ ions.

(3) Decomposition of ozone in presence of nitrogen monoxide.

(4) Combination between dinitrogen and dihydrogen to form ammonia in the presence of finely divided iron.

58 The given compound



is an example of _____.

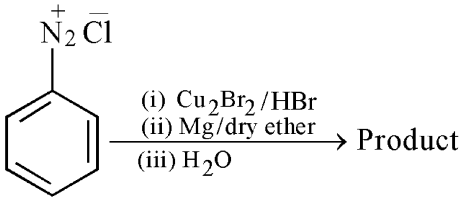
(1) benzylic halide

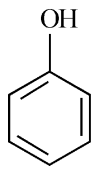
(2) aryl halide

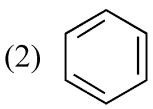
(3) allylic halide

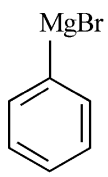
(4) vinylic halide

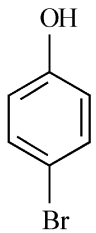
59 Identify the product in the following reaction:



(1) 

(2) 

(3) 

(4) 

60 Given below are two statements : one is labelled as **Assertion A** and the other is labelled as **Reason R** :

Assertion A : Helium is used to dilute oxygen in diving apparatus.

Reasons R : Helium has high solubility in O_2.

In the light of the above statements, choose the **correct** answer from the options given below :

(1) Both **A** and **R** are true and **R** is the correct explanation of **A**.

(2) Both **A** and **R** are true and **R** is **NOT** the correct explanation of **A**.

(3) **A** is true but **R** is false.

(4) **A** is false but **R** is true.

61 A compound is formed by two elements A and B. The element B forms cubic close packed structure and atoms of A occupy 1/3 of tetrahedral voids. If the formula of the compound is A_xB_y, then the value of $x + y$ is in option

(1) 5 (2) 4

(3) 3 (4) 2

62 Given below are two statements :

Statement I : A unit formed by the attachment of a base to 1' position of sugar is known as nucleoside

Statement II : When nucleoside is linked to phosphorous acid at 5'-position of sugar moiety, we get nucleotide.

In the light of the above statements, choose the **correct** answer from the options given below :

(1) Both Statement **I** and Statement **II** are true.

(2) Both Statement **I** and Statement **II** are false.

(3) Statement **I** is true but Statement **II** is false.

(4) Statement **I** is false but Statement **II** is true.

63 The relation between n_m, (n_m = the number of permissible values of magnetic quantum number (m)) for a given value of azimuthal quantum number (l), is

(1) $l = \frac{n_m - 1}{2}$ (2) $l = 2n_m + 1$

(3) $n_m = 2l^2 + 1$ (4) $n_m = l + 2$

64 Amongst the following, the total number of species NOT having eight electrons around central atom in its outer most shell, is

$NH_3, AlCl_3, BeCl_2, CCl_4, PCl_5$:

(1) 3 (2) 2

(3) 4 (4) 1

65 The **correct** order of energies of molecular orbitals of N_2 molecule, is :

(1) $\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < \left(\pi 2p_x = \pi 2p_y\right) < \sigma 2p_z < \left(\pi^* 2p_x = \pi^* 2p_y\right) < \sigma^* 2p_z$

(2) $\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < \sigma 2p_z < \left(\pi 2p_x = \pi 2p_y\right) < \left(\pi^* 2p_x = \pi^* 2p_y\right) < \sigma^* 2p_z$

(3) $\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < \sigma 2p_z < \sigma^* 2p_z < \left(\pi 2p_x = \pi 2p_y\right) < \left(\pi^* 2p_x = \pi^* 2p_y\right)$

(4) $\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < \left(\pi 2p_x = \pi 2p_y\right) < \left(\pi^* 2p_x = \pi^* 2p_y\right) < \sigma 2p_z < \sigma^* 2p_z$

66 The number of σ bonds, π bonds and lone pair of electrons in pyridine, respectively are:

(1) 11, 2, 0 (2) 12, 3, 0

(3) 11, 3, 1 (4) 12, 2, 1

67 Intermolecular forces are forces of attraction and repulsion between interacting particles that will include :

A. dipole - dipole forces.

B. dipole - induced dipole forces.

C. hydrogen bonding.

D. covalent bonding.

E. dispersion forces.

Choose the **most appropriate** answer from the options given below :

(1) B, C, D, E are correct.

(2) A, B, C, D are correct.

(3) A, B, C, E are correct.

(4) A, C, D, E are correct.

68 Which of the following statements are **NOT** correct?

A. Hydrogen is used to reduce heavy metal oxides to metals.

B. Heavy water is used to study reaction mechanism.

C. Hydrogen is used to make saturated fats from oils.

D. The H-H bond dissociation enthalpy is lowest as compared to a single bond between two atoms of any element.

E. Hydrogen reduces oxides of metals that are more active than iron.

Choose the **most appropriate** answer from the options given below :

(1) B, C, D, E only

(2) B, D only

(3) D, E only

(4) A, B, C only

69 Which amongst the following molecules on polymerization produces neoprene?

(1) $H_2C = CH - CH = CH_2$

(2) $H_2C = \underset{\displaystyle Cl}{\underset{|}{C}} - CH = CH_2$

(3) $H_2C = CH - C \equiv CH$

(4) $H_2C = \underset{\displaystyle CH_3}{\underset{|}{C}} - CH = CH_2$

70 Some tranquilizers are listed below. Which one from the following belongs to barbiturates?

(1) Chlordiazepoxide
(2) Meprobamate
(3) Valium
(4) Veronal

71 The element expected to form largest ion to achieve the nearest noble gas configuration is :

(1) O (2) F
(3) N (4) Na

72 Select the **correct** statements from the following :

A. Atoms of all elements are composed of two fundamental particles.
B. The mass of the electron is 9.10939×10^{-31} kg.
C. All the isotopes of a given element show same chemical properties.
D. Protons and electrons are collectively known as nucleons.
E. Dalton's atomic theory, regarded the atom as an ultimate particle of matter.

Choose the **correct** answer from the options given below :

(1) A, B and C only
(2) C, D and E only
(3) A and E only
(4) B, C and E only

73 Consider the following reaction and identify the product (P).

$$\underset{\displaystyle CH_3}{\underset{|}{CH}} - \underset{\displaystyle OH}{\underset{|}{CH}} - CH_3 \xrightarrow{HBr} \text{Product (P)}$$

3 – Methylbutan – 2 – ol

(1) $CH_3 - \underset{\displaystyle CH_3}{\underset{|}{\overset{\displaystyle Br}{\overset{|}{C}}}} - CH_2 - CH_3$

(2) $CH_3\,CH = CH - CH_3$

(3) $CH_3 - \underset{\displaystyle CH_3}{\underset{|}{CH}} - \underset{\displaystyle Br}{\underset{|}{CH}} - CH_3$

(4) $CH_3 - \underset{\displaystyle CH_3}{\underset{|}{\overset{\displaystyle CH_3}{\overset{|}{C}}}} - CH_2\,Br$

74 The stability of Cu^{2+} is more than Cu^{+} salts in aqueous solution due to -

(1) first ionisation enthalpy.
(2) enthalpy of atomization.
(3) hydration energy.
(4) second ionisation enthalpy.

75 Which one of the following statements is **correct**?

(1) The daily requirement of Mg and Ca in the human body is estimated to be 0.2 - 0.3 g.
(2) All enzymes that utilise ATP in phosphate transfer require Ca as the cofactor.
(3) The bone in human body is an inert and unchanging substance.
(4) Mg plays roles in neuromuscular function and interneuronal transmission.

76 Weight (g) of two moles of the organic compound, which is obtained by heating sodium ethanoate with sodium hydroxide in presence of calcium oxide is :

(1) 16 (2) 32

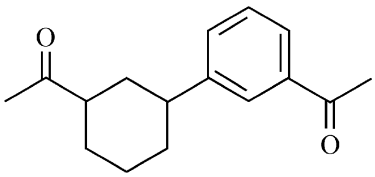
(3) 30 (4) 18

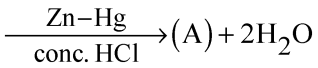
77 Amongst the given options which of the following molecules / ion acts as a Lewis acid?

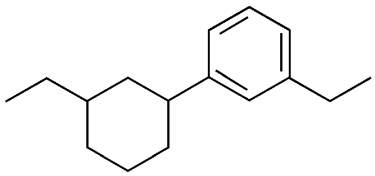
(1) NH_3 (2) H_2O

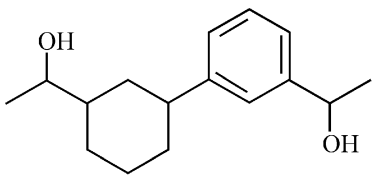
(3) BF_3 (4) OH^-

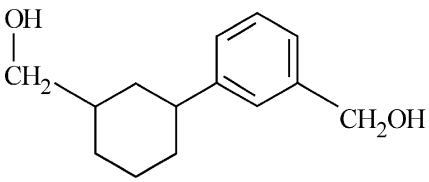
78 Identify product (A) in the following reaction:

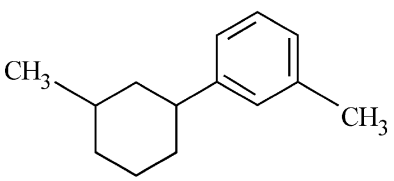




(1) 

(2) 

(3) 

(4) 

79 Taking stability as the factor, which one of the following represents **correct** relationship?

(1) $TlCl_3 > TlCl$

(2) $InI_3 > InI$

(3) $AlCl > AlCl_3$

(4) $TlI > TlI_3$

80 Homoleptic complex from the following complexes is :

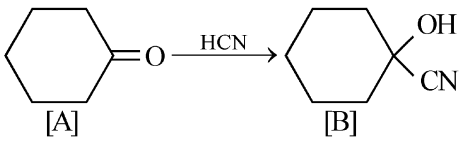
(1) Potassium trioxalatoaluminate (III)

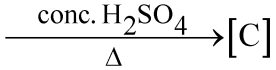
(2) Diamminechloridonitrito - N - platinum (II)

(3) Pentaamminecarbonatocobalt (III) chloride

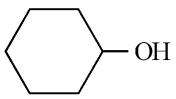
(4) Triamminetriaquachromium (III) chloride

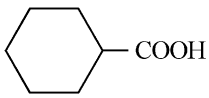
81 Complete the following reaction :

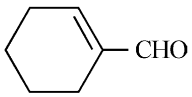


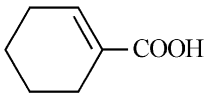


[C] is _____.

(1) 

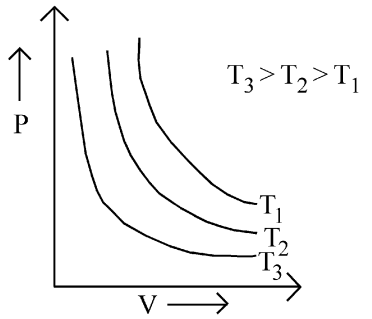
(2) 

(3) 

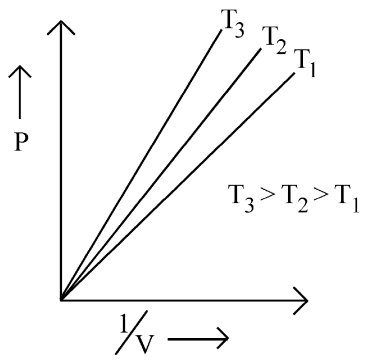
(4) 

82 Which amongst the following options is **correct** graphical representation of Boyle's Law?

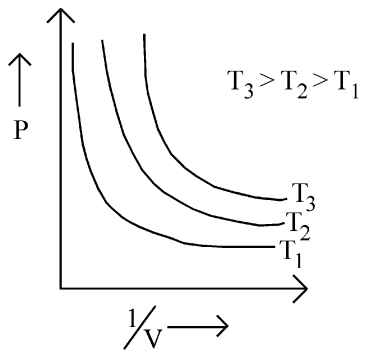
(1)



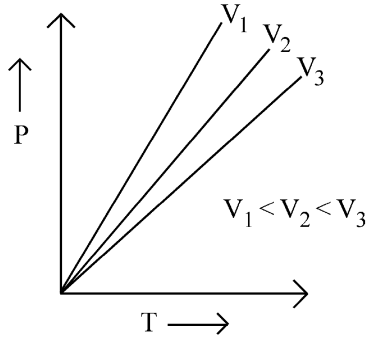
(2)



(3)



(4)



83 The **right** option for the mass of CO_2 produced by heating 20 g of 20% pure limestone is (Atomic mass of Ca = 40)

$$\left[CaCO_3 \xrightarrow{1200\ K} CaO + CO_2\right]$$

(1) 1.12 g (2) 1.76 g

(3) 2.64 g (4) 1.32 g

84 For a certain reaction, the rate = $k[A]^2[B]$, when the initial concentration of A is tripled keeping concentration of B constant, the initial rate would

(1) decrease by a factor of nine.

(2) increase by a factor of six.

(3) increase by a factor of nine.

(4) increase by a factor of three.

85 Given below are two statements : one is labelled as **Assertion A** and the other is labelled as **Reason R** :

Assertion A : In equation $\Delta_r G = -nFE_{cell}$, value of $\Delta_r G$ depends on n.

Reasons R : E_{cell} is an intensive property and $\Delta_r G$ is an extensive property.

In the light of the above statements, choose the **correct** answer from the options given below :

(1) Both **A** and **R** are true and **R** is the correct explanation of **A**.

(2) Both **A** and **R** are true and **R** is **NOT** the correct explanation of **A**.

(3) **A** is true but **R** is false.

(4) **A** is false but **R** is true.

Chemistry : Section-B (Q. No. 86 to 100)

86 Match **List - I** with **List - II** :

List - I (Oxoacids of Sulphur)	**List - II (Bonds)**
A. Peroxodisulphuric acid	I. Two S-OH, Four S=O, One S-O-S
B. Sulphuric acid	II. Two S-OH, One S=O
C. Pyrosulphuric acid	III. Two S-OH, Four S=O, One S-O-O-S
D. Sulphurous acid	IV. Two S-OH, Two S=O

Choose the **correct** answer from the options given below :

(1) A-I, B-III, C-II, D-IV

(2) A-III, B-IV, C-I, D-II

(3) A-I, B-III, C-IV, D-II

(4) A-III, B-IV, C-II, D-I

87 Which of the following statements are **INCORRECT**?

A. All the transition metals except scandium form MO oxides which are ionic.

B. The highest oxidation number corresponding to the group number in transition metal oxides is attained in Sc_2O_3 to Mn_2O_7.

C. Basic character increases from V_2O_3 to V_2O_4 to V_2O_5.

D. V_2O_4 dissolves in acids to give VO_4^{3-} salts.

E. CrO is basic but Cr_2O_3 is amphoteric.

Choose the **correct** answer from the options given below :

(1) A and E only

(2) B and D only

(3) C and D only

(4) B and C only

88 Which complex compound is most stable?

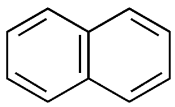
(1) $\left[Co(NH_3)_4(H_2O)Br\right](NO_3)_2$

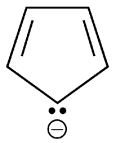
(2) $\left[Co(NH_3)_3(NO_3)_3\right]$

(3) $\left[CoCl_2(en)_2\right]NO_3$

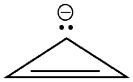
(4) $\left[Co(NH_3)_6\right]_2(SO_4)_3$

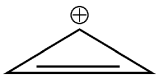
89 Consider the following compounds/species:

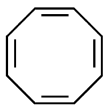
i. 

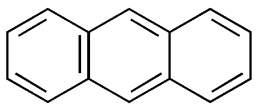
ii. 

iii. 

iv. 

v. 

vi. 

vii. 

The number of compounds/species which obey Huckel's rule is _____.

(1) 4 (2) 6

(3) 2 (4) 5

90 What fraction of one edge centred octahedral void lies in one unit cell of fcc?

(1) $\frac{1}{2}$ (2) $\frac{1}{3}$

(3) $\frac{1}{4}$ (4) $\frac{1}{12}$

91 Which amongst the following options is the **correct** relation between change in enthalpy and change in internal energy?

(1) $\Delta H = \Delta U - \Delta n_g RT$

(2) $\Delta H = \Delta U + \Delta n_g RT$

(3) $\Delta H - \Delta U = -\Delta nRT$

(4) $\Delta H + \Delta U = \Delta nR$

92 On balancing the given redox reaction,

$$a\,Cr_2O_7^{2-} + b\,SO_3^{2-}(aq) + c\,H^+(aq) \rightarrow$$

$$2a\,Cr^{3+}(aq) + b\,SO_4^{2-}(aq) + \frac{c}{2}H_2O(\ell)$$

the coefficients a, b and c are found to be, respectively -

(1) 1, 3, 8 (2) 3, 8, 1

(3) 1, 8, 3 (4) 8, 1, 3

93 The equilibrium concentrations of the species in the reaction $A + B \rightleftharpoons C + D$ are 2, 3, 10 and 6 mol L^{-1}, respectively at 300 K. $\Delta G°$ for the reaction is (R = 2 cal / mol K)

(1) 1372.60 cal (2) – 137.26 cal

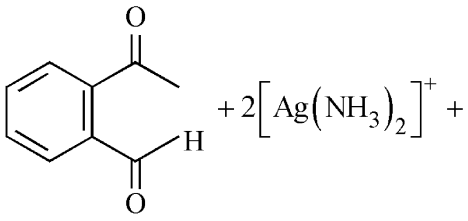
(3) – 1381.80 cal (4) – 13.73 cal

94 Pumice stone is an example of -

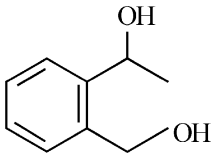
(1) sol (2) gel

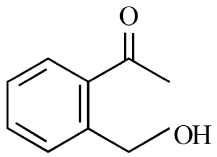
(3) solid sol (4) foam

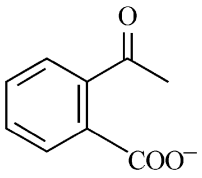
95 Identify the major product obtained in the following reaction :

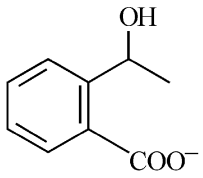


$$3\,^-OH \xrightarrow{\Delta} \text{major product}$$

(1) 

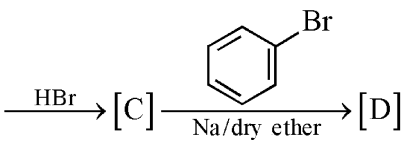
(2) 

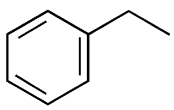
(3) 

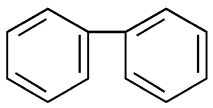
(4) 

96 Identify the final product [D] obtained in the following sequence of reactions.

$$CH_3CHO \xrightarrow[ii)\ H_3O^+]{i)\ LiAlH_4} [A] \xrightarrow[\Delta]{H_2SO_4} [B]$$



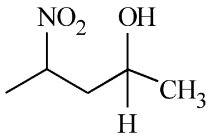
(1) 

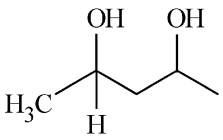
(2) 

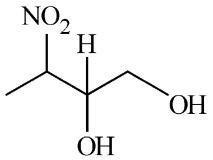
(3) C_4H_{10}

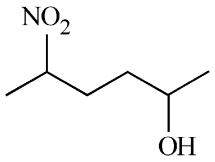
(4) $HC \equiv C^{\ominus}\ Na^+$

97 Which amongst the following will be most readily dehydrated under acidic conditions ?

(1) 

(2) 

(3) 

(4) 

98 Given below are two statements :

Statement I : The nutrient deficient water bodies lead to eutrophication.

Statement II : Eutrophication leads to decrease in the level of oxygen in the water bodies.

In the light of the above statements, choose the **correct** answer from the options given below :

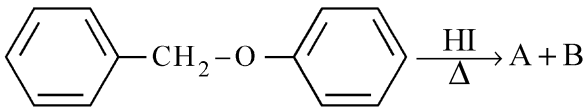
(1) Both **Statement I** and **Statement II** are true.

(2) Both **Statement I** and **Statement II** are false.

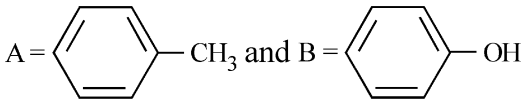
(3) **Statement I** is correct but **Statement II** is false.

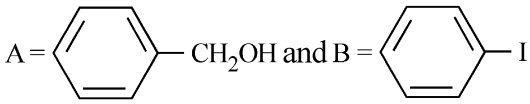
(4) **Statement I** is incorrect but **Statement II** is true.

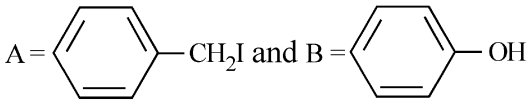
99 Consider the following reaction :

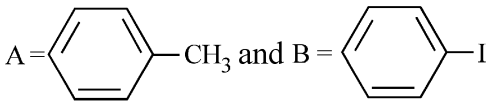


Identify products A and B.

(1) 

(2) 

(3) 

(4) 

100 The reaction that does **NOT** take place in a blast furnace between 900 K to 1500 K temperature range during extraction of iron is :

(1) $Fe_2O_3 + CO \rightarrow 2FeO + CO_2$

(2) $FeO + CO \rightarrow Fe + CO_2$

(3) $C + CO_2 \rightarrow 2CO$

(4) $CaO + SiO_2 \rightarrow CaSiO_3$

Botany : Section-A (Q. No. 101 to 135)

101 Given below are two statements : One is labelled as **Assertion A** and the other is labelled as **Reason R** :
Assertion A : The first stage of gametophyte in the life cycle of moss is protonema stage.
Reason R : Protonema develops directly from spores produced in capsule.
In the light of the above statements, choose the **most appropriate** answer from the options given below :
(1) Both **A** and **R** are correct and **R** is the correct explanation of **A**.
(2) Both **A** and **R** are correct but **R** is NOT the correct explanation of **A**.
(3) **A** is correct but **R** is not correct.
(4) **A** is not correct but **R** is correct.

102 In angiosperm, the haploid, diploid and triploid structures of a fertilized embryo sac sequentially are :
(1) Synergids, Primary endosperm nucleus and zygote
(2) Antipodals, synergids, and primary endosperm nucleus
(3) Synergids, Zygote and Primary endosperm nucleus
(4) Synergids, antipodals and Polar nuclei

103 Movement and accumulation of ions across a membrane against their concentration gradient can be explained by
(1) Osmosis
(2) Facilitated Diffusion
(3) Passive Transport
(4) Active Transport

104 Large, colourful, fragrant flowers with nectar are seen in :
(1) insect pollinated plants
(2) bird pollinated plants
(3) bat pollinated plants
(4) wind pollinated plants

105 The phenomenon of pleiotropism refers to
(1) presence of several alleles of a single gene controlling a single crossover.
(2) presence of two alleles, each of the two genes controlling a single trait.
(3) a single gene affecting multiple phenotypic expression.
(4) more than two genes affecting a single character.

106 Which hormone promotes internode/petiole elongation in deep water rice?
(1) GA_3 (2) Kinetin
(3) Ethylene (4) 2, 4-D

107 Among 'The Evil Quartet', which one is considered the most important cause driving extinction of species?
(1) Habitat loss and fragmentation
(2) Over exploitation for economic gain
(3) Alien species invasions
(4) Co-extinctions

108 Upon exposure to UV radiation, DNA stained with ethidium bromide will show
(1) Bright red colour
(2) Bright blue colour
(3) Bright yellow colour
(4) Bright orange colour

109 Which micronutrient is required for splitting of water molecule during photosynthesis?
(1) manganese (2) molybdenum
(3) magnesium (4) copper

110 Axile placentation is observed in
(1) Mustard, Cucumber and Primrose
(2) China rose, Beans and Lupin
(3) Tomato, Dianthus and Pea
(4) China rose, Petunia and Lemon

111 The process of appearance of recombination nodules occurs at which sub stage of prophase I in meiosis?
(1) Zygotene (2) Pachytene
(3) Diplotene (4) Diakinesis

112 The reaction centre in PS II has an absorption maxima at
(1) 680 nm (2) 700 nm
(3) 660 nm (4) 780 nm

113 Unequivocal proof that DNA is the genetic material was first proposed by
(1) Frederick Griffith
(2) Alfred Hershey and Martha Chase
(3) Avery, Macleoid and McCarthy
(4) Wilkins and Franklin